

Akshay Agrawal

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PROFILE

Full-stack software developer with two years of production experience at a German software house, delivering public-sector and B2B products in Java/Spring Boot and TypeScript/Angular. Core contributor on a statewide identity-management platform and on AI-assisted products, including a customer-support chatbot and a Microsoft 365 assistant built with LangChain and RAG. Comfortable across the stack from PostgreSQL and message queues to Angular, Vaadin, and Ionic mobile apps, with a strong testing habit and daily use of AI coding tools such as Claude Code and GitHub Copilot.

WORK EXPERIENCE

PCT digital GmbH, Kiel, Germany — **Full-Stack Software Developer**

Sept 2024 – Ongoing

Identity and access management platform for a state-level public-sector IT provider, education sector, statewide rollout (May 2026 – present)

- Core contributor on a Java 25 / Spring Boot 4 system that consolidates teacher and student data from several source systems and provisions school accounts to a central directory service.
- Implemented ingest, merge, and provisioning features within a hexagonal architecture, keeping business logic in a framework-free domain module behind explicit ports and adapters.
- Built the API-key-secured ingest REST API, source-priority merge rules, and a transactional outbox with RabbitMQ acknowledgements for reliable asynchronous provisioning.
- Implemented entitlement-engine rules for role assignment and account lifecycle, role-based access control from Keycloak token claims, and account name and password generation.
- Delivered internal Vaadin UI features, including searchable and exportable persona views, functional mailbox management, school-admin administration, a dashboard, and a demo simulator.
- Maintained quality gates: GitHub Actions pipelines, Testcontainers integration tests, Liquibase migrations, and an OpenAPI contract test between API and frontend. Regular code reviewer on the team.

Waste-management customer app and portal for a municipal-services provider, white-labelled for regional utilities (Oct 2024 – Jan 2025, Oct 2025 – May 2026)

- Built a customer-support AI chatbot end-to-end: a LangChain agent with feature-specific tools for collection calendars, invoices, payments, bulky-waste orders, and service tickets, backed by retrieval over FAQ content in PostgreSQL pgvector.
- Designed guardrails and operations for the chatbot, including persisted chat history, rate limiting, message-size limits, modular system prompts, and AI-driven mass test campaigns to validate answer quality.
- Enabled the assistant to pre-fill and steer the bulky-waste order form, connecting the conversational layer to the Angular UI.
- Implemented the digital customer card feature across backend, frontend, data sync, and a mock external API.
- Remediated penetration-test findings such as account takeover, account enumeration, insecure direct object references, authorization bypass, SQL injection, and XSS.
- Shipped features and fixes for the Ionic/Capacitor iOS and Android app and the Angular portal, including the admin statistics area, bin-change flows, receipt filtering performance and accessibility, with Cypress end-to-end coverage.

Science-and-industry matching platform for a regional development agency (Feb – May 2025)

- Delivered profile, catalogue and multi-step project registration features on an Nx monorepo with Angular, tRPC, Express, Knex and PostgreSQL, plus dark mode, responsive layouts and end-to-end scenarios.
- Hardened authentication and data integrity: JWT expiry handling, Keycloak single-sign-on settings, ownership and permission checks, duplicate prevention and matching edge cases.

AI office assistant for Microsoft 365 with Node.js, tRPC, Prisma, pgvector, LangChain/Mistral and Bot Framework (Jun – Sept 2025)

- Lead developer of the Microsoft Teams bot and a main backend contributor for an LLM assistant that reads and drafts emails and manages calendar events via Microsoft Graph.

- Implemented intent detection and query analysis ahead of vector search, an email draft-and-confirm conversation flow, calendar event creation and conversation-history prompting with robust LLM output parsing.
- Built scheduled morning notifications with server-sent-event subscriptions persisted in the backend, Graph permission scopes and German localisation.

Christian-Albrechts-Universität zu Kiel — Research Assistant, Information Systems Apr 2023 – Sept 2023

- Analysed the university's computer-science course offerings to surface patterns and correlations that informed curriculum development.
- Modelled course relationships in a Neo4j graph database and built a Flask web interface for interactive exploration and visualisation.

ZeroBS GmbH, Kiel, Germany — Full-Stack Developer (Working Student) Oct 2021 – July 2022

- Developed a CMS platform with a Flask backend and JavaScript/Jinja2 frontend that tripled content-team productivity.
- Automated manual operational processes and introduced one-click deployments, cutting operational overhead by around ninety percent and giving the client full deployment autonomy.
- Managed cloud infrastructure on AWS and DigitalOcean with Terraform and maintained asynchronous job queues with Celery and RabbitMQ.

EDUCATION

Christian-Albrechts-Universität zu Kiel, Germany 2019 – 2023

Master of Science (M.Sc.) | Major: Computer Science

Xavier Institute of Engineering, Mumbai, India 2015 – 2019

Bachelor of Engineering (B.E.) | Major: Computer Science

SKILLS & INTERESTS

- **Languages:** TypeScript, Java, JavaScript, Python, SQL, HTML/CSS
- **Backend:** Spring Boot, Node.js, Express, tRPC, ts-rest, Prisma, Knex, Liquibase, REST and OpenAPI, RabbitMQ, hexagonal architecture
- **Frontend and mobile:** Angular, Vaadin, Ionic, Capacitor (iOS/Android), Tailwind CSS, ngx-formly, Nx monorepos
- **AI engineering:** LangChain, retrieval-augmented generation (pgvector, Pinecone), tool-calling agents, prompt design, Mistral and OpenAI APIs, AI-driven test automation
- **Databases:** PostgreSQL, pgvector, SQLite, Neo4j
- **Security and identity:** Keycloak, OAuth2/OIDC, JWT, Microsoft Graph and MSAL, role-based access control, penetration-test remediation
- **DevOps:** Docker, GitHub Actions, GitLab CI, Kubernetes (k3d, Helm, Tilt), OpenTofu/Terraform, Ansible, Open Telekom Cloud, Testcontainers
- **Testing:** JUnit, Jest, Vitest, Cypress/Cucumber, test-driven development
- **AI tooling:** Claude Code, GitHub Copilot
- **Languages spoken:** English (fluent), German (intermediate, B1/B2), Hindi (native)
- **Interests:** Photography, swimming, cooking, volleyball